SUPPORTING ADULT GUIDANCE

Loci — the shape of a rule

Constructions, Loci and Bearings · Lesson 5. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

This lesson names and generalises the **locus** — the set of all points satisfying a rule — and is the capstone of the perpendicular- and angle-bisector work. Learners meet the two everyday loci (a fixed distance from a point, a fixed distance from a line) and the key GCSE skill of combining loci to identify a region. It leans on discovery (tracing equal-distance points until a shape appears) before formal construction.

Driving Question: *What shape do all the points obeying a rule make?*

The mathematics, in plain terms

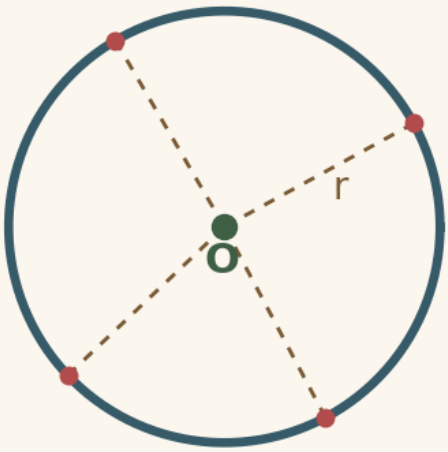
A locus is simply 'all the points where a rule is true'. The rule 'a fixed distance from a point' gives a circle; 'a fixed distance from a straight segment' gives two parallel lines joined by semicircular ends (the ends are arcs because, beyond an endpoint, the nearest part of the segment is that endpoint). Region problems combine rules: draw each rule's boundary (a circle, or a bisector) and shade the part satisfying all of them at once — the overlap. The two bisectors from earlier lessons are themselves loci: equidistant from two points, and equidistant from two lines.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — The locus a fixed distance from a point

Draw the locus of all points exactly 3 cm from a point O.

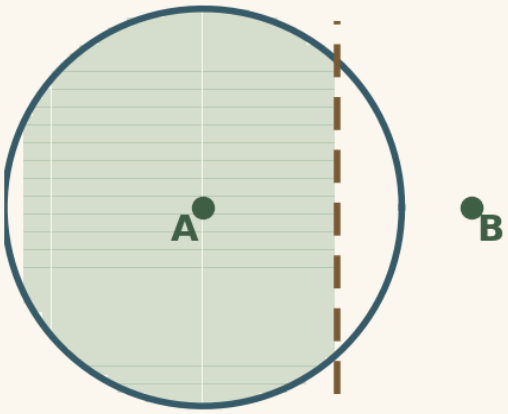


1. Mark the point O.	point O
2. Every point of the locus is 3 cm from O.	distance 3 cm
3. Set the compass to 3 cm and draw the circle.	radius 3 cm

The locus is a circle of radius 3 cm, centred O — every point on it is 3 cm from O.

Example 2 — A region from two rules

Shade the points that are within 3 cm of A AND closer to A than to B.



1. 'Within 3 cm of A' — draw a circle of radius 3 cm centred A; inside it obeys the rule.	circle at A
2. 'Closer to A than B' — construct the perpendicular bisector of AB; A's side obeys the rule.	perp. bisector
3. Shade where both are true — inside the circle and on A's side.	overlap

The region is the part of the circle on A's side of the perpendicular bisector — where the two loci overlap.

Discover the shape before naming it

Let learners collect equal-distance points and watch a shape emerge before the word 'locus' or the construction arrives. Meeting a circle as 'all the points a fixed distance from here' — rather than a shape to memorise — is the whole gain. For region problems, insist on drawing each boundary first and reasoning about which side satisfies the rule, then shading; the shading is the conclusion, not the starting point.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	Change a goat's rope length and watch its circular grazing region grow — a fixed distance from a point.	~9 min

Movement	Pick a rule and drag a point; the marks reveal the locus (a circle, or a rounded racetrack).	~13 min
Reflection	See why a fixed distance from a segment has semicircular ends (arcs about the endpoints).	~9 min
Creativity	Combine two rules and shade the region where they overlap.	~9 min
Rest	A short pause to notice loci in the everyday world.	~2 min
Nutrition	Name what the learning feeds: defining a shape by a rule, and the usefulness of loci.	~3 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

“A fixed distance from a point is just one point, or a straight line.”

It is a whole circle. Ask the learner to find three different points the same distance from O, then more, and see them fall on a circle.

“A fixed distance from a line is a single parallel line.”

There are two parallel lines (one each side) and, for a finite segment, semicircular ends. Ask what the nearest point of the segment is when you go past its end.

“The region is the whole circle / the whole line.”

A region problem needs every rule at once. Ask the learner to test a point in their shaded area against each rule in turn.

Shading before drawing the boundaries.

Draw each locus first, then reason which side or part satisfies the rule. Ask: 'What is the boundary of this rule, and which side obeys it?'

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the perpendicular bisector (Lesson 3) and the angle bisector (Lesson 4). A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Suppose a learner says the locus a fixed distance from a point is 'a dot' or 'a line', or shades a whole circle for a two-rule region. The aim is a question that returns the reasoning to them.

- “Find three points exactly this far from O. Now three more. What shape are they falling on?”
- “Go past the end of the segment — what is the nearest point of it now? So what shape is that end?”
- “Pick a point in your shaded region. Does it obey BOTH rules? Now pick one just outside.”
- “What is the boundary of each rule here, and which side of it satisfies the rule?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

Is 'loci' just the plural of 'locus'?
Yes — one locus, several loci. Both are worth using so learners recognise the exam wording.

Do learners need to construct the racetrack precisely?
Understanding its shape (parallel sides, semicircular ends) matters most. Precise construction of the parallel parts comes with the perpendicular tools; the ends are arcs about the endpoints.

How is a region problem marked?
Look for each boundary constructed correctly, correct reasoning about which side/part obeys each rule, and the overlap clearly shaded.

How does this connect to the earlier lessons?
The perpendicular bisector and angle bisector are loci too — equidistant from two points, and from two lines. This lesson gives them their proper name and adds the distance loci.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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